

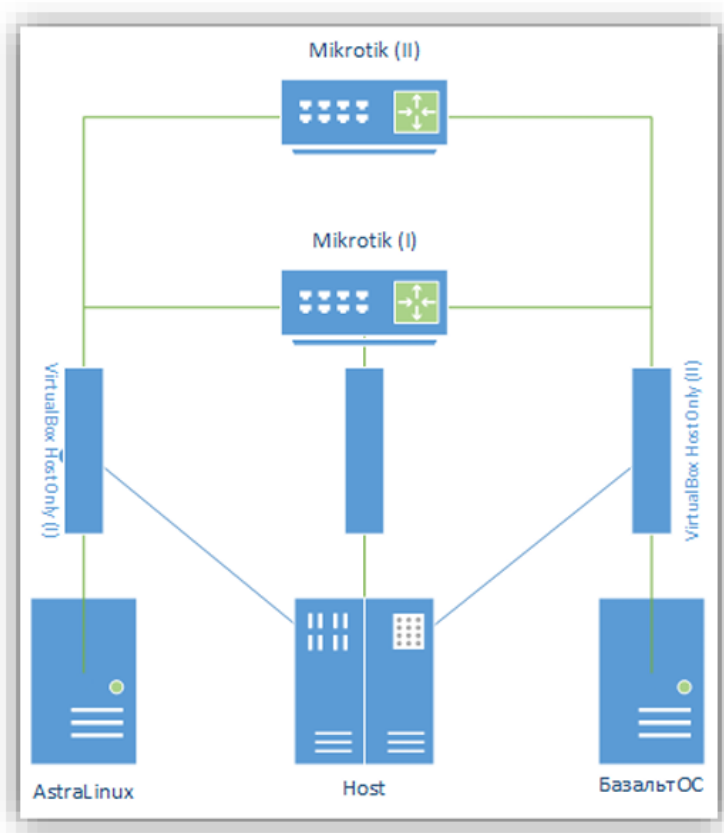
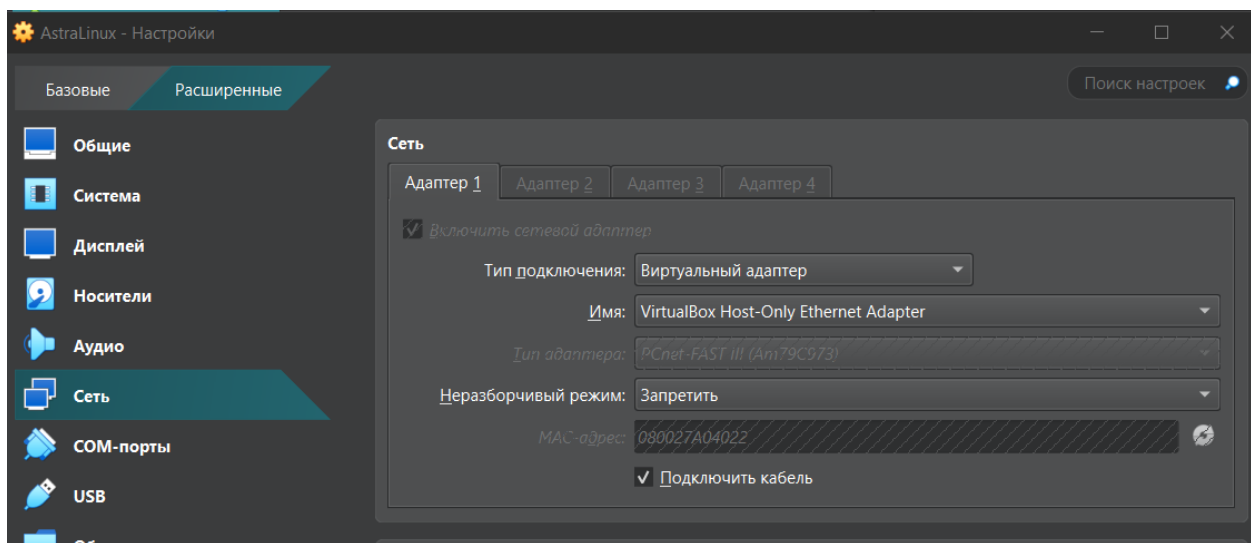


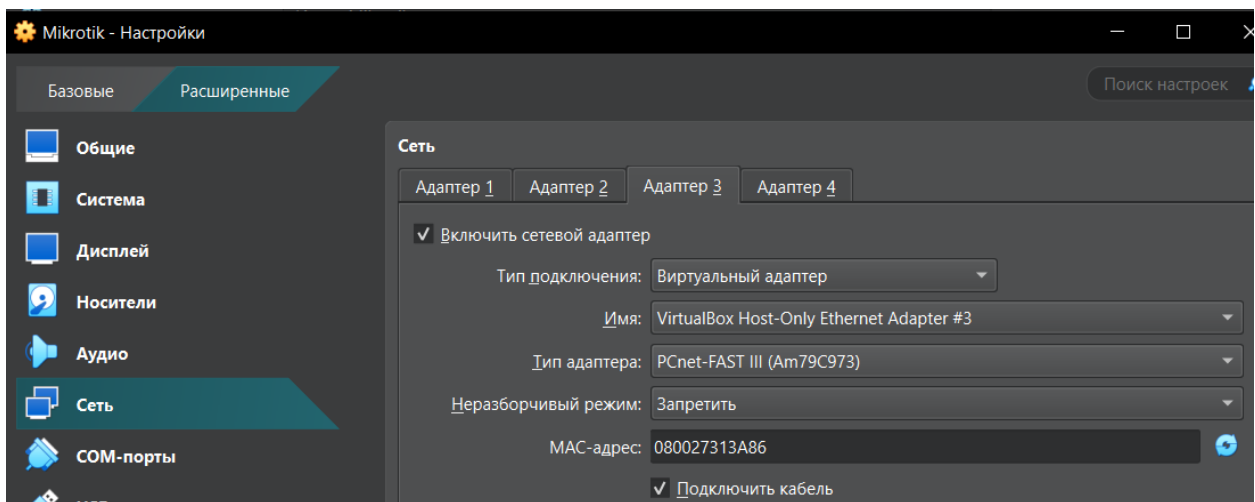
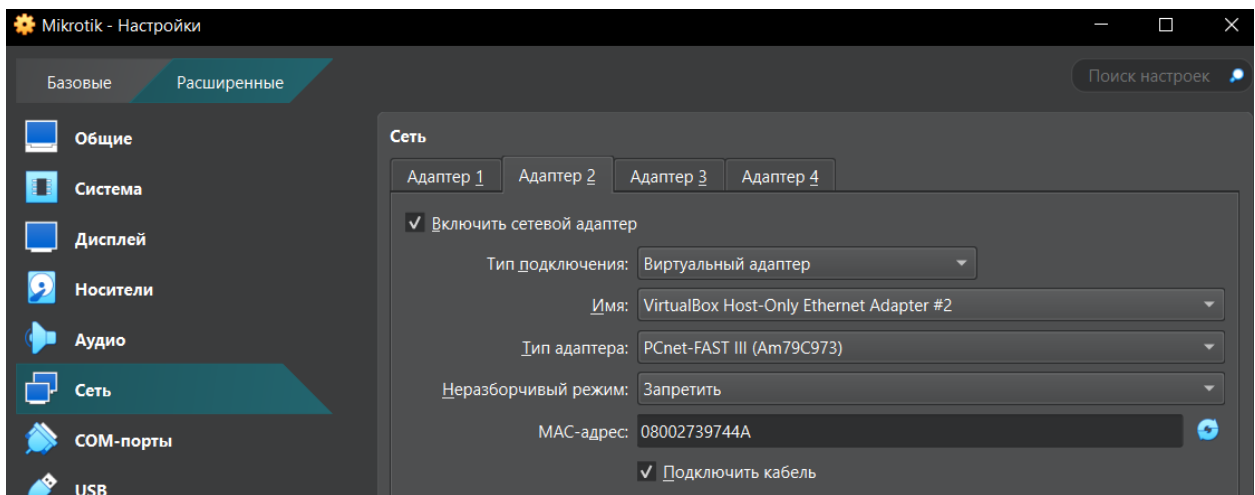
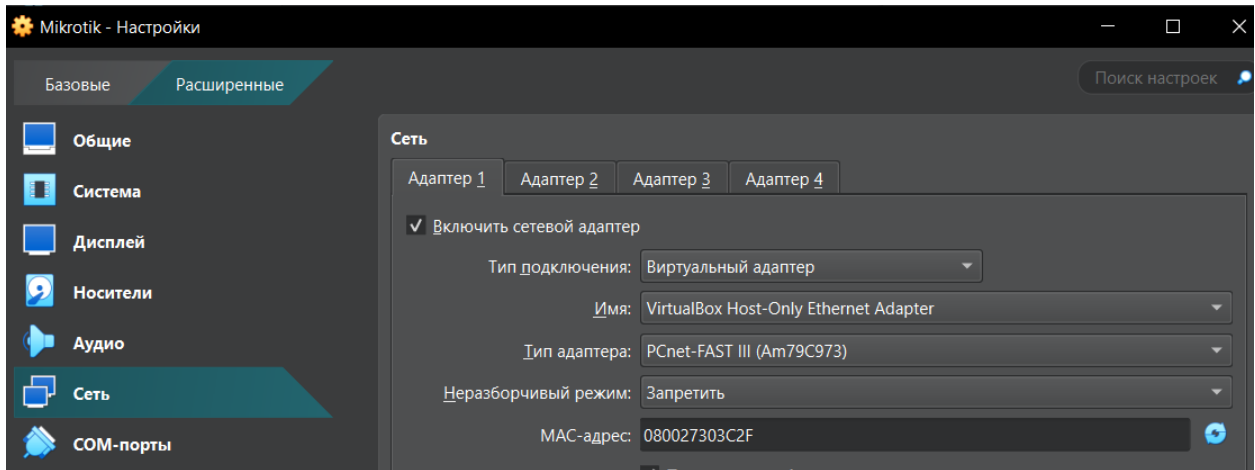
Рисунок 1 – Конфигурация сети для практического занятия

Конфигурация:

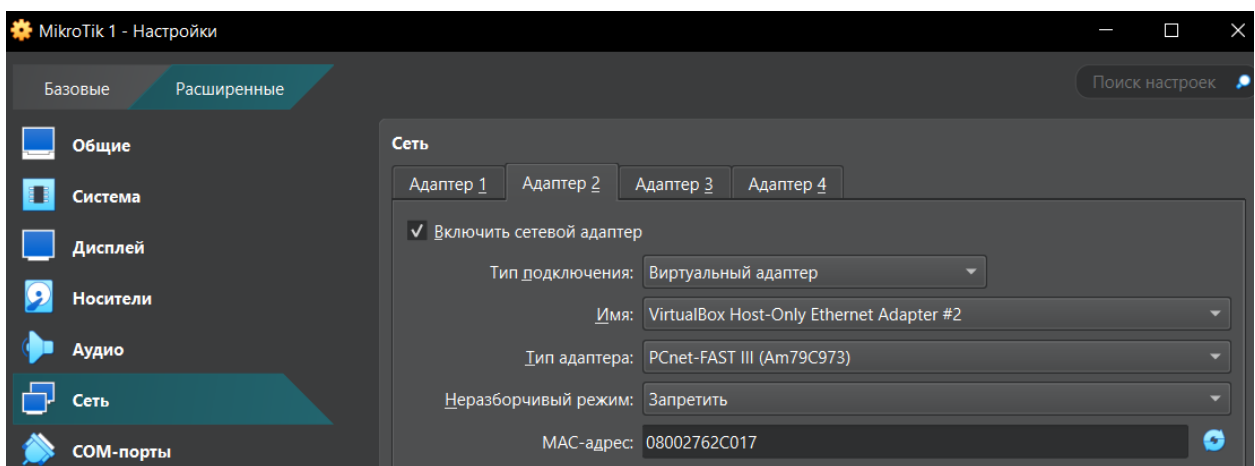
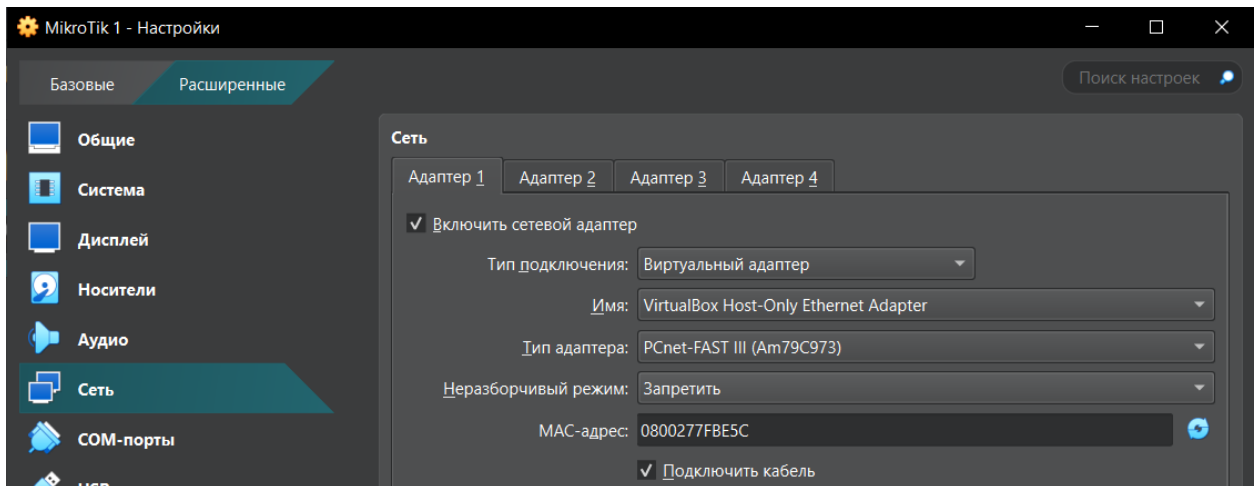
Астра Линукс:



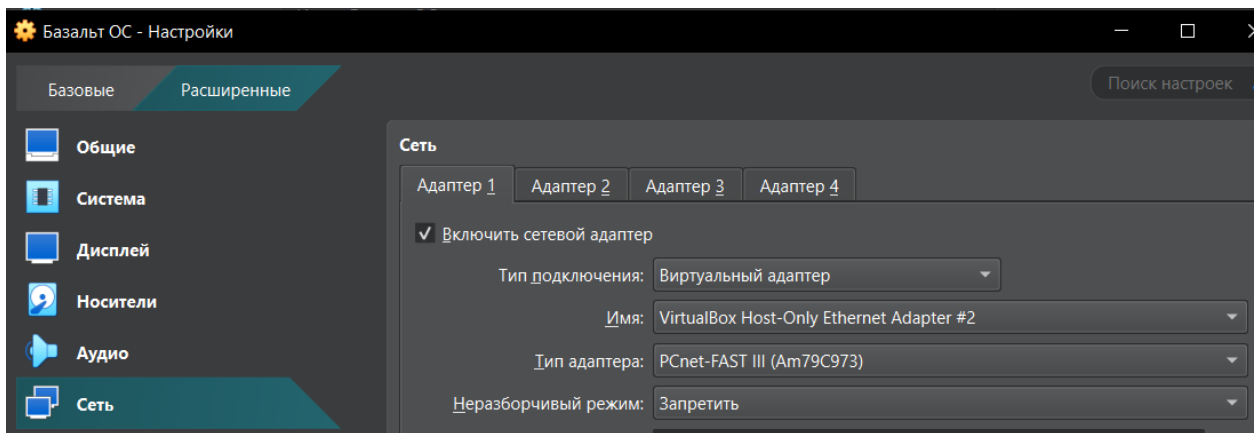
Микротик:



Микротик 1:



БазальтОС:



1.

Исходные данные:

Подсеть 10.10.N.0/24

Разделение подсети:

Первая подсеть: 10.10.16.0/25 (адреса 10.10.16.1–10.10.16.126).

Вторая подсеть: 10.10.16.128/25 (адреса 10.10.16.129–10.10.16.254)
255.255.255.128 (/25)

VirtualBox Host Only 1: 10.10.16.1/25

VirtualBox Host Only 2: 10.10.16.129/25

Имя	IPv4 префикс	IPv6 префикс	DHCP сервер
VirtualBox Host-Only Ethernet Adapter	10.10.16.1/25		Выключен
VirtualBox Host-Only Ethernet Adapter #2	10.10.16.129/25		Выключен
VirtualBox Host-Only Ethernet Adapter #3	192.168.19.1/24		Выключен

Астра Линукс:

```
root@host-astra:~ # cat /etc/network/interfaces.d/eth0
auto eth0
iface eth0 inet static
    address 10.10.16.5
    network 255.255.255.128
    gateway 10.10.16.1

root@host-astra:~ # sudo ip addr add 10.10.16.5/25 dev eth0
root@host-astra:~ # ip a | grep "eth0"
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN group default qlen 1000
    inet 10.10.16.5/25 scope global eth0
root@host-astra:~ #
```

Микротик:

10.10.16.2/25 (ether1)

10.10.16.130/25 (ether2)

```
[admin@MikroTik1] > ip address/add address=10.10.16.2/25
interface: ether1
[admin@MikroTik1] > ip address/add address=10.10.16.130/25
interface: ether2
[admin@MikroTik1] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS      NETWORK      INTERFACE
0 10.10.16.2/25  10.10.16.0   ether1
1 10.10.16.130/25 10.10.16.128 ether2
[admin@MikroTik1] >
```

Микротик 1:

10.10.16.3/25 (ether2)

10.10.16.131/25 (ether3)

```
[admin@MikroTik1] > ip address/add address=10.10.16.3/25
interface: ether2
[admin@MikroTik1] > ip address/add address=10.10.16.131/25
interface: ether3
[admin@MikroTik1] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 10.10.16.3/25 10.10.16.0 ether2
1 10.10.16.131/25 10.10.16.128 ether3
[admin@MikroTik1] >
```

БазальтОС:

10.10.23.132/25 (enp0s3)

```
[root@host-basalt ~]# sudo ip addr add 10.10.16.132/25 dev enp0s3
[root@host-basalt ~]# ip a | grep "enp0s3"
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default
    inet 10.16.63.253/18 brd 10.16.63.255 scope global dynamic noprefixroute enp0s3
    inet 10.10.16.132/25 scope global enp0s3
[root@host-basalt ~]# ip a | grep "scope global enp0s3"
    inet 10.10.16.132/25 scope global enp0s3
[root@host-basalt ~]#
```

Проверка пинга (должен проходить только между адресами, находящимися в одной подсети):

Астра Линукс <=> Микротик:

```
root@host-astara:~# ping 10.10.16.2
PING 10.10.16.2 (10.10.16.2) 56(84) bytes of data:
64 bytes from 10.10.16.2: icmp_seq=1 ttl=64 time=0.922 ms
64 bytes from 10.10.16.2: icmp_seq=2 ttl=64 time=1.14 ms
64 bytes from 10.10.16.2: icmp_seq=3 ttl=64 time=1.51 ms
64 bytes from 10.10.16.2: icmp_seq=4 ttl=64 time=1.09 ms
64 bytes from 10.10.16.2: icmp_seq=5 ttl=64 time=2.47 ms
64 bytes from 10.10.16.2: icmp_seq=6 ttl=64 time=2.22 ms
64 bytes from 10.10.16.2: icmp_seq=7 ttl=64 time=1.18 ms
64 bytes from 10.10.16.2: icmp_seq=8 ttl=64 time=1.51 ms
64 bytes from 10.10.16.2: icmp_seq=9 ttl=64 time=1.82 ms
64 bytes from 10.10.16.2: icmp_seq=10 ttl=64 time=2.13 ms
^C
--- 10.10.16.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9020ms
rtt min/avg/max/ndev = 0.922/1.603/2.479/0.510 ms
root@host-astara:~#
```

```
[admin@MikroTik1] > ping 10.10.16.5
3.252 - lease expired
SEQ HOST SIZE TTL TIME STATUS
0 10.10.16.5 56 64 3ms31us
1 10.10.16.5 56 64 3ms37us
2 10.10.16.5 56 64 2ms58us
3 10.10.16.5 56 64 2ms363us
4 10.10.16.5 56 64 859us
5 10.10.16.5 56 64 1ms872us
6 10.10.16.5 56 64 3ms55us
7 10.10.16.5 56 64 1ms806us
8 10.10.16.5 56 64 928us
9 10.10.16.5 56 64 1ms669us
10 10.10.16.5 56 64 2ms18us
sent=11 received=11 packet-loss=0% min-rtt=859us avg-rtt=2ms63us
max-rtt=3ms55us
```

Астра Линукс <=> Микротик 1:

```

root@host-astra:~# ping 10.10.16.3
PING 10.10.16.3 (10.10.16.3) 56(84) bytes of data.
64 bytes from 10.10.16.3: icmp_seq=1 ttl=64 time=2.07 ms
64 bytes from 10.10.16.3: icmp_seq=2 ttl=64 time=2.46 ms
64 bytes from 10.10.16.3: icmp_seq=3 ttl=64 time=2.08 ms
64 bytes from 10.10.16.3: icmp_seq=4 ttl=64 time=1.89 ms
64 bytes from 10.10.16.3: icmp_seq=5 ttl=64 time=2.06 ms
64 bytes from 10.10.16.3: icmp_seq=6 ttl=64 time=2.06 ms
^C
--- 10.10.16.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/ndev = 1.893/2.110/2.468/0.181 ms
root@host-astra:~#

[admin@MikroTik1] > ping 10.10.16.5
SEQ HOST                                SIZE TTL TIME                        STATUS
0 10.10.16.5                            56 64 2ms771us
1 10.10.16.5                            56 64 2ms397us
2 10.10.16.5                            56 64 1ms862us
3 10.10.16.5                            56 64 1ms588us
4 10.10.16.5                            56 64 2ms528us
5 10.10.16.5                            56 64 2ms51us
6 10.10.16.5                            56 64 2ms7us
sent=7 received=7 packet-loss=0% min-rtt=1ms588us avg-rtt=2ms172us
max-rtt=2ms771us

[admin@MikroTik1] >

```

Микротик <=> Микротик 1:

```

[admin@MikroTik1] > ping 10.10.16.3
SEQ HOST                                SIZE TTL TIME                        STATUS
0 10.10.16.3                            56 64 1ms486us
1 10.10.16.3                            56 64 2ms501us
2 10.10.16.3                            56 64 2ms13us
3 10.10.16.3                            56 64 1ms747us
4 10.10.16.3                            56 64 2ms149us
5 10.10.16.3                            56 64 1ms233us
6 10.10.16.3                            56 64 2ms526us
sent=7 received=7 packet-loss=0% min-rtt=1ms233us avg-rtt=1ms950us
max-rtt=2ms526us

[admin@MikroTik1] > ping 10.10.16.2
SEQ HOST                                SIZE TTL TIME                        STATUS
0 10.10.16.2                            56 64 3ms203us
1 10.10.16.2                            56 64 2ms879us
2 10.10.16.2                            56 64 1ms951us
3 10.10.16.2                            56 64 1ms808us
4 10.10.16.2                            56 64 2ms117us
5 10.10.16.2                            56 64 2ms883us
6 10.10.16.2                            56 64 2ms22us
sent=7 received=7 packet-loss=0% min-rtt=1ms808us avg-rtt=2ms409us
max-rtt=3ms203us

```

Пинг не проходит в разных подсетях.

БазальтОС <=> Микротик:

```

root@host-basalt:~# ping 10.10.16.130
PING 10.10.16.130 (10.10.16.130) 56(84) bytes of data.
64 bytes from 10.10.16.130: icmp_seq=1 ttl=64 time=2.72 ms
64 bytes from 10.10.16.130: icmp_seq=2 ttl=64 time=2.36 ms
64 bytes from 10.10.16.130: icmp_seq=3 ttl=64 time=1.61 ms
64 bytes from 10.10.16.130: icmp_seq=4 ttl=64 time=1.93 ms
64 bytes from 10.10.16.130: icmp_seq=5 ttl=64 time=2.29 ms
^C
--- 10.10.16.130 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4811ms
rtt min/avg/max/ndev = 1.613/2.182/2.717/0.379 ms

[admin@MikroTik1] > ping 10.10.16.132
SEQ HOST                                SIZE TTL TIME                        STATUS
0 10.10.16.132                          56 64 2ms378us
1 10.10.16.132                          56 64 2ms278us
2 10.10.16.132                          56 64 3ms228us
3 10.10.16.132                          56 64 2ms459us
4 10.10.16.132                          56 64 3ms17us
sent=5 received=5 packet-loss=0% min-rtt=2ms278us avg-rtt=2ms670us
max-rtt=3ms228us

[admin@MikroTik1] >

```

БазальтОС <=> Микротик 1:

```

root@host-basalt:~# ping 10.10.16.131
PING 10.10.16.131 (10.10.16.131) 56(84) bytes of data.
64 bytes from 10.10.16.131: icmp_seq=1 ttl=64 time=3.47 ms
64 bytes from 10.10.16.131: icmp_seq=2 ttl=64 time=1.90 ms
64 bytes from 10.10.16.131: icmp_seq=3 ttl=64 time=1.11 ms
64 bytes from 10.10.16.131: icmp_seq=4 ttl=64 time=1.96 ms
64 bytes from 10.10.16.131: icmp_seq=5 ttl=64 time=2.94 ms
^C
--- 10.10.16.131 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/ndev = 1.114/2.276/3.474/0.832 ms
root@host-basalt:~#

[admin@MikroTik1] > ping 10.10.16.132
interface: port snmp user ping
SEQ HOST                                SIZE TTL TIME                        STATUS
0 10.10.16.132                          56 64 1ms308us
1 10.10.16.132                          56 64 1ms766us
2 10.10.16.132                          56 64 2ms928us
3 10.10.16.132                          56 64 2ms88us
4 10.10.16.132                          56 64 2ms428us
5 10.10.16.132                          56 64 2ms618us
sent=6 received=6 packet-loss=0% min-rtt=1ms308us avg-rtt=2ms189us
max-rtt=2ms928us

```

```

[root@host-basalt ~]# ping 10.10.16.2
PING 10.10.16.2 (10.10.16.2) 56(84) bytes of data.
From 10.16.63.253 icmp_seq=1 Destination Host Unreachable
From 10.16.63.253 icmp_seq=2 Destination Host Unreachable
From 10.16.63.253 icmp_seq=3 Destination Host Unreachable
^C
--- 10.10.16.2 ping statistics ---
5 packets transmitted, 0 received, +3 errors, 100% packet loss, time 4069ms
pipe 4
[root@host-basalt ~]# ping 10.10.16.3
PING 10.10.16.3 (10.10.16.3) 56(84) bytes of data.
From 10.16.63.253 icmp_seq=1 Destination Host Unreachable
From 10.16.63.253 icmp_seq=2 Destination Host Unreachable
From 10.16.63.253 icmp_seq=3 Destination Host Unreachable
^C
--- 10.10.16.3 ping statistics ---
4 packets transmitted, 0 received, +3 errors, 100% packet loss, time 3076ms
pipe 4
[root@host-basalt ~]# ping 10.10.16.5
PING 10.10.16.5 (10.10.16.5) 56(84) bytes of data.
From 10.16.63.253 icmp_seq=1 Destination Host Unreachable
From 10.16.63.253 icmp_seq=2 Destination Host Unreachable
From 10.16.63.253 icmp_seq=3 Destination Host Unreachable
^C
--- 10.10.16.5 ping statistics ---
4 packets transmitted, 0 received, +3 errors, 100% packet loss, time 3055ms
pipe 4
[root@host-basalt ~]# _

```

Астра пингуется только с первыми интерфейсами Микротиков, Базальт со вторыми интерфейсами Микротиков и у Микротиков пингуются так первый-первый, второй-второй.

2. Создание моста:

```

[admin@MikroTik1] > interface/bridge/add name=bridge1
[admin@MikroTik1] > interface/bridge/print
Flags: X - disabled, R - running
 0 R name="bridge1" mtu=auto actual-mtu=1500 l2mtu=65535 arp=enabled
    arp-timeout=auto mac-address=36:3E:F7:08:6A:66 protocol-mode=rstp
    fast-forward=yes igmp-snooping=no auto-mac=yes ageing-time=5m
    priority=0x8000 max-message-age=20s forward-delay=15s
    transmit-hold-count=6 vlan-filtering=no dhcp-snooping=no
    port-cost-mode=long mvrp=no max-learned-entries=auto
[admin@MikroTik1] > _

```

Добавление интерфейсов в мост (через ВебФиг):

```
[admin@MikroTik1] > interface/bridge/port print
Columns: INTERFACE, BRIDGE, HW, PVID, PRIORITY, HORIZON
# INTERFACE BRIDGE HW PVID PRIORITY HORIZON
0 ether1 bridge1 yes 1 0x80 none
1 ether2 bridge1 yes 1 0x80 none
[admin@MikroTik1] >
```

- **INTERFACE:** Имя интерфейса (ether1, ether2).
- **BRIDGE:** Имя моста (bridge1).
- **HW:** yes — аппаратная поддержка моста (корректно).
- **PVID:** Уникальный идентификатор порта (1 для обоих — нормально).
- **PRIORITY:** Приоритет порта (0x00 — значение по умолчанию).
- **HORIZON:** none — горизонт петли не настроен (по умолчанию).

```
[admin@MikroTik1] > ip address/print
Flags: S - SLAVE
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 S 10.10.16.2/25 10.10.16.0 ether1
1 S 10.10.16.130/25 10.10.16.128 ether2
[admin@MikroTik1] > ip address/remove
numbers: 0,1
[admin@MikroTik1] > ip address/add address=10.10.16.2/25
interface: bridge1
[admin@MikroTik1] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS NETWORK INTERFACE
0 10.10.16.2/25 10.10.16.0 bridge1
[admin@MikroTik1] >
```

3.

```
[admin@MikroTik1] > interface/vlan/add name=vlan2 interface=bridge1
vlan-id: 2
[admin@MikroTik1] > interface/bridge/set bridge1 vlan-filtering=yes
[admin@MikroTik1] > interface/bridge/vlan/add bridge=bridge1 vlan-ids=2 tagged=ether1
[admin@MikroTik1] > interface/bridge/vlan/add bridge=bridge1 vlan-ids=2 tagged=ether2
[admin@MikroTik1] > interface/bridge/vlan/add bridge=bridge1 vlan-ids=1 tagged=ether2
[admin@MikroTik1] > interface/bridge/vlan/add bridge=bridge1 vlan-ids=1 tagged=ether1
[admin@MikroTik1] > ip address/add address=10.10.16.129/25
interface: vlan2
[admin@MikroTik1] >
```

Тегирование vlan2 проходит на VirtualBox Host Only 1 и VirtualBox Host Only 2 (Астра – VirtualBox Host Only 1 (Ethernet), Базальт – VirtualBox Host Only 2 (Ethernet 3)). В VirtualBox Host Only 3 нет тегированного трафика.

4. Назначение IP для виртуального VLAN-интерфейса Микротика:

```
[admin@MikroTik1] > ip address/add address=10.10.16.200/25
interface: vlan2
[admin@MikroTik1] > interface/print
Flags: R - RUNNING; S - SLAVE
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAC-ADDRESS
#  NAME      TYPE      ACTUAL-MTU  L2MTU  MAC-ADDRESS
0  RS ether1   ether      1500      08:00:27:30:3C:2F
1  RS ether2   ether      1500      08:00:27:39:74:4A
2  R  ether3   ether      1500      08:00:27:31:3A:86
3  R  bridge1  bridge     1500     65535    08:00:27:30:3C:2F
4  R  lo       loopback   65536     00:00:00:00:00:00
5  R  vlan2    vlan       1500     65531    08:00:27:30:3C:2F
[admin@MikroTik1] > ip address/print
Columns: ADDRESS, NETWORK, INTERFACE
#  ADDRESS      NETWORK      INTERFACE
0  10.10.16.2/25  10.10.16.0   bridge1
1  10.10.16.129/25 10.10.16.128 vlan2
2  10.10.16.200/25 10.10.16.128 vlan2
[admin@MikroTik1] >
```

Astra Linux – настройка VLAN 2

Создание VLAN-интерфейса на eth0:

```

root@host-astra:~ # sudo ip link add link eth0 name eth0.2 type vlan id 2
root@host-astra:~ # ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN group default qlen 1000
    link/ether 08:00:27:a0:40:22 brd ff:ff:ff:ff:ff:ff
    inet 10.10.16.5/25 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fd00:2005:5:8000:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591230sec preferred_lft 604030sec
    inet6 fd00:2005:5:23:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591220sec preferred_lft 604020sec
    inet6 fd00:2005:5:4000:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591920sec preferred_lft 604720sec
    inet6 fd00:2005:5:1000:a00:27ff:fea0:4022/64 scope global mngtmpaddr dynamic
        valid_lft 2591220sec preferred_lft 604020sec
    inet6 fe80::a00:27ff:fea0:4022/64 scope link
        valid_lft forever preferred_lft forever
3: eth0.2@eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 08:00:27:a0:40:22 brd ff:ff:ff:ff:ff:ff
root@host-astra:~ #

```

```

root@host-astra:~ # sudo ip addr add 10.10.16.201/25 dev eth0.2
root@host-astra:~ # sudo ip link set eth0.2 up
root@host-astra:~ # ip a | grep eth0.2
3: eth0.2@eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    inet 10.10.16.201/25 scope global eth0.2
root@host-astra:~ # _

```

```

root@host-astra:~ # ping 10.10.16.200
PING 10.10.16.200 (10.10.16.200) 56(84) bytes of data.
64 bytes from 10.10.16.200: icmp_seq=1 ttl=64 time=4.24 ms
64 bytes from 10.10.16.200: icmp_seq=2 ttl=64 time=2.21 ms
64 bytes from 10.10.16.200: icmp_seq=3 ttl=64 time=2.61 ms
64 bytes from 10.10.16.200: icmp_seq=4 ttl=64 time=2.23 ms
^C
--- 10.10.16.200 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3011ms
rtt min/avg/max/mdev = 2.212/2.827/4.241/0.832 ms
root@host-astra:~ #

```

Проверка прохода пинга с интерфейса eth0.2:

```

root@host-astra:~ # ping 10.10.16.200 -I eth0.2
PING 10.10.16.200 (10.10.16.200) from 10.10.16.201 eth0.2: 56(84) bytes of data.
64 bytes from 10.10.16.200: icmp_seq=1 ttl=64 time=1.64 ms
64 bytes from 10.10.16.200: icmp_seq=2 ttl=64 time=3.86 ms
64 bytes from 10.10.16.200: icmp_seq=3 ttl=64 time=1.88 ms
64 bytes from 10.10.16.200: icmp_seq=4 ttl=64 time=3.25 ms
^C
--- 10.10.16.200 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3043ms
rtt min/avg/max/mdev = 1.649/2.664/3.867/0.927 ms
root@host-astra:~ # _

```

Пинг на vlan2 проходит с eth0.2.